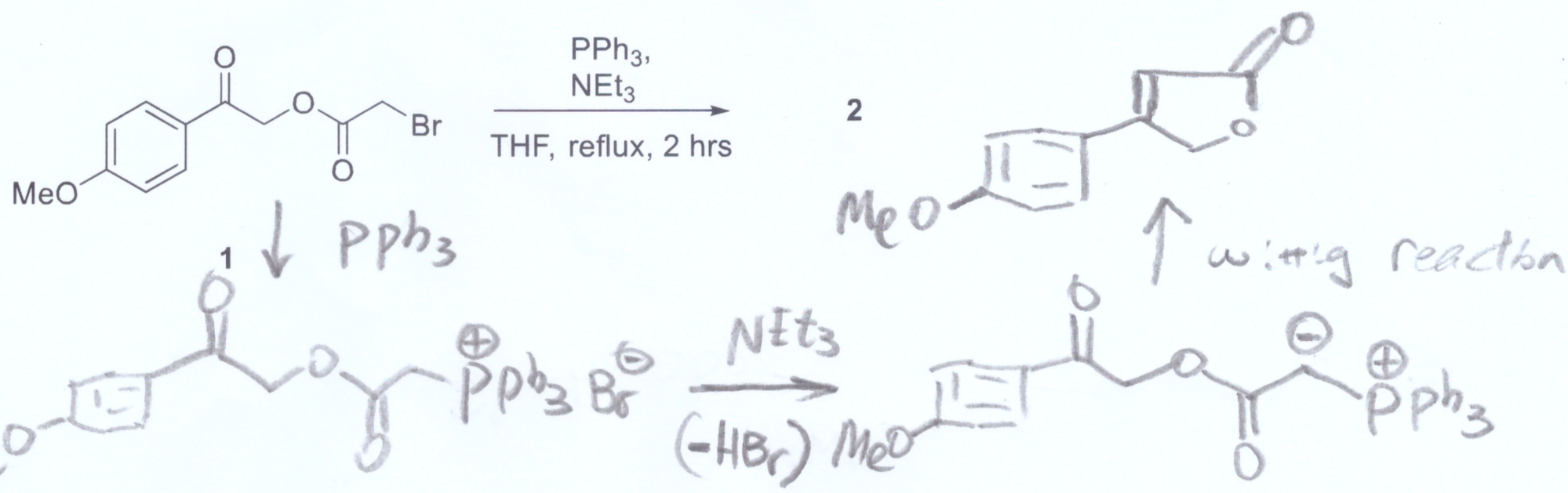
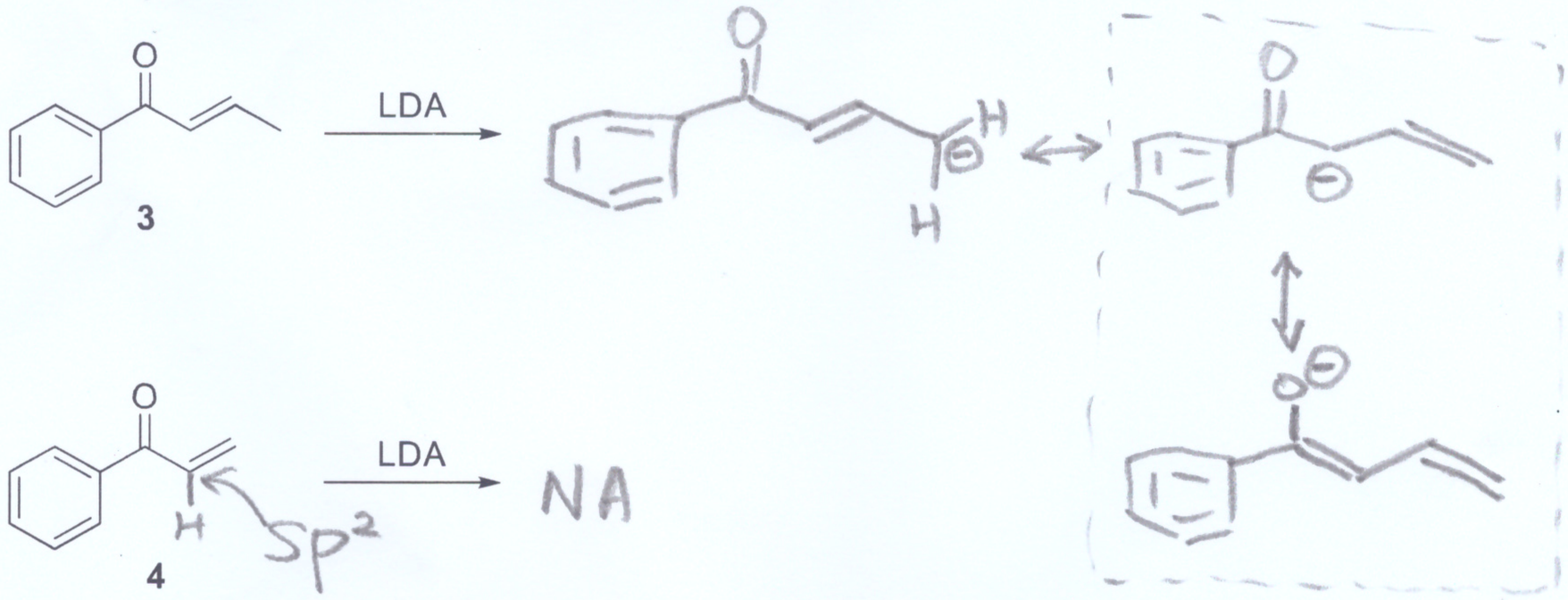
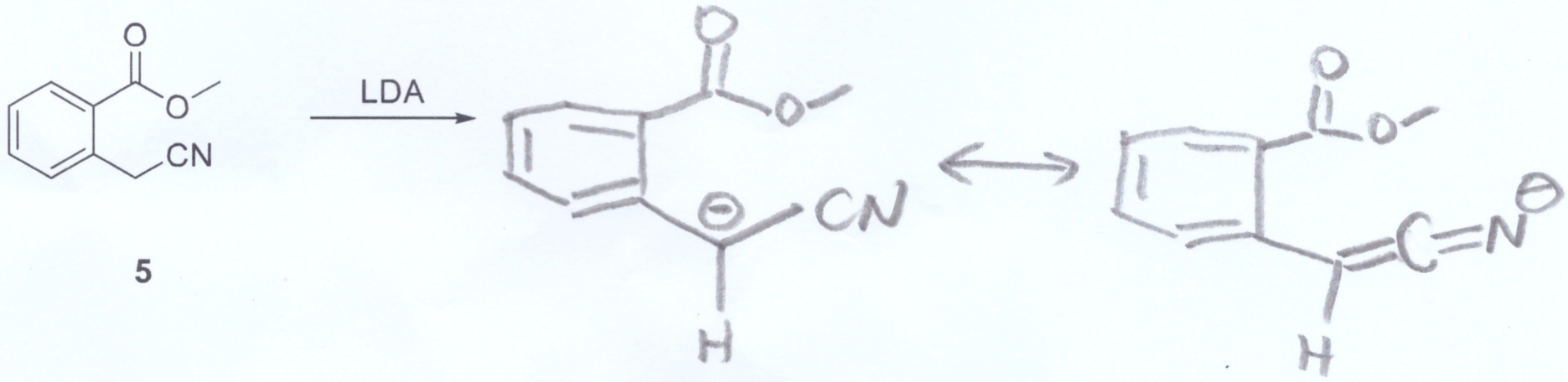
Question 1: Provide the reaction intermediates and products for the following reaction



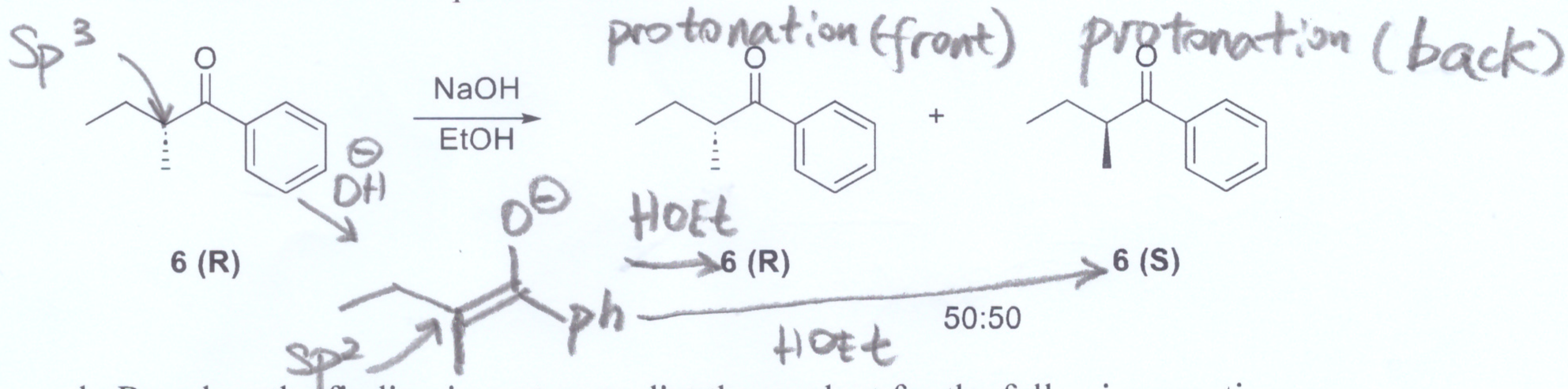
Question 2: Provide the corresponding enolate from the following compounds, if no enolate formed, place NA in the answer.



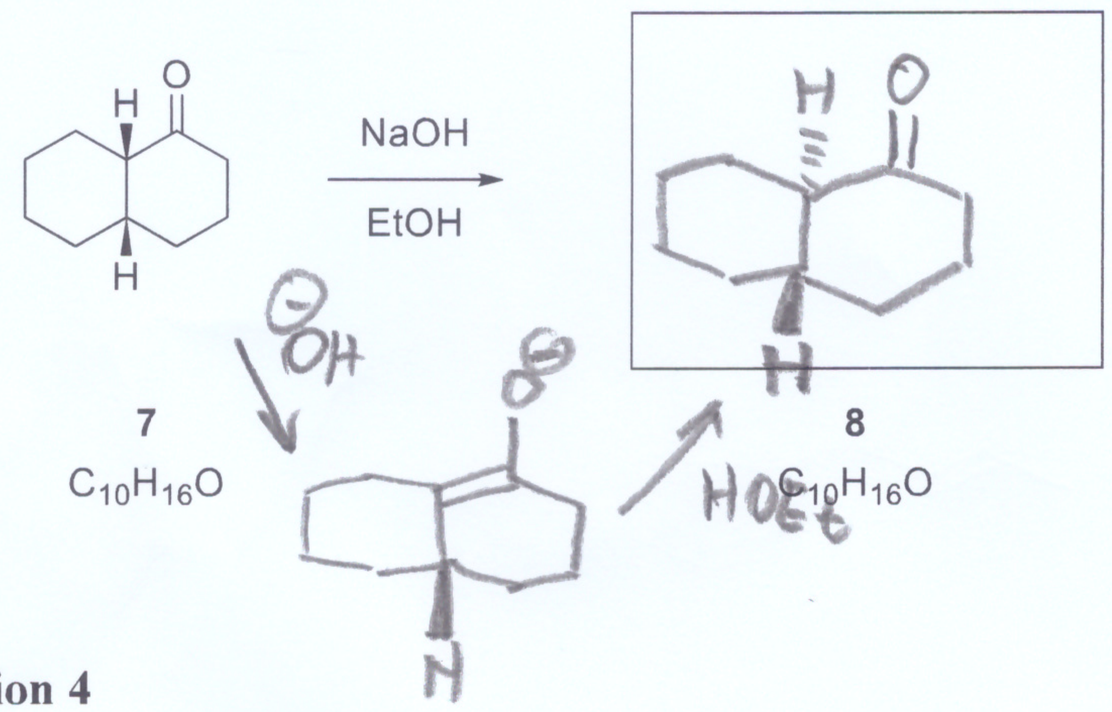


Question 3

a. When a solution of (R)-(+)-2-methyl-1-phenylbutan-1-one in aqueous ethanol is treated with a base, the solution gradually loses its optical activity. Isolation of the ketones shows it has been racemized. Propose a mechanism for this transformation

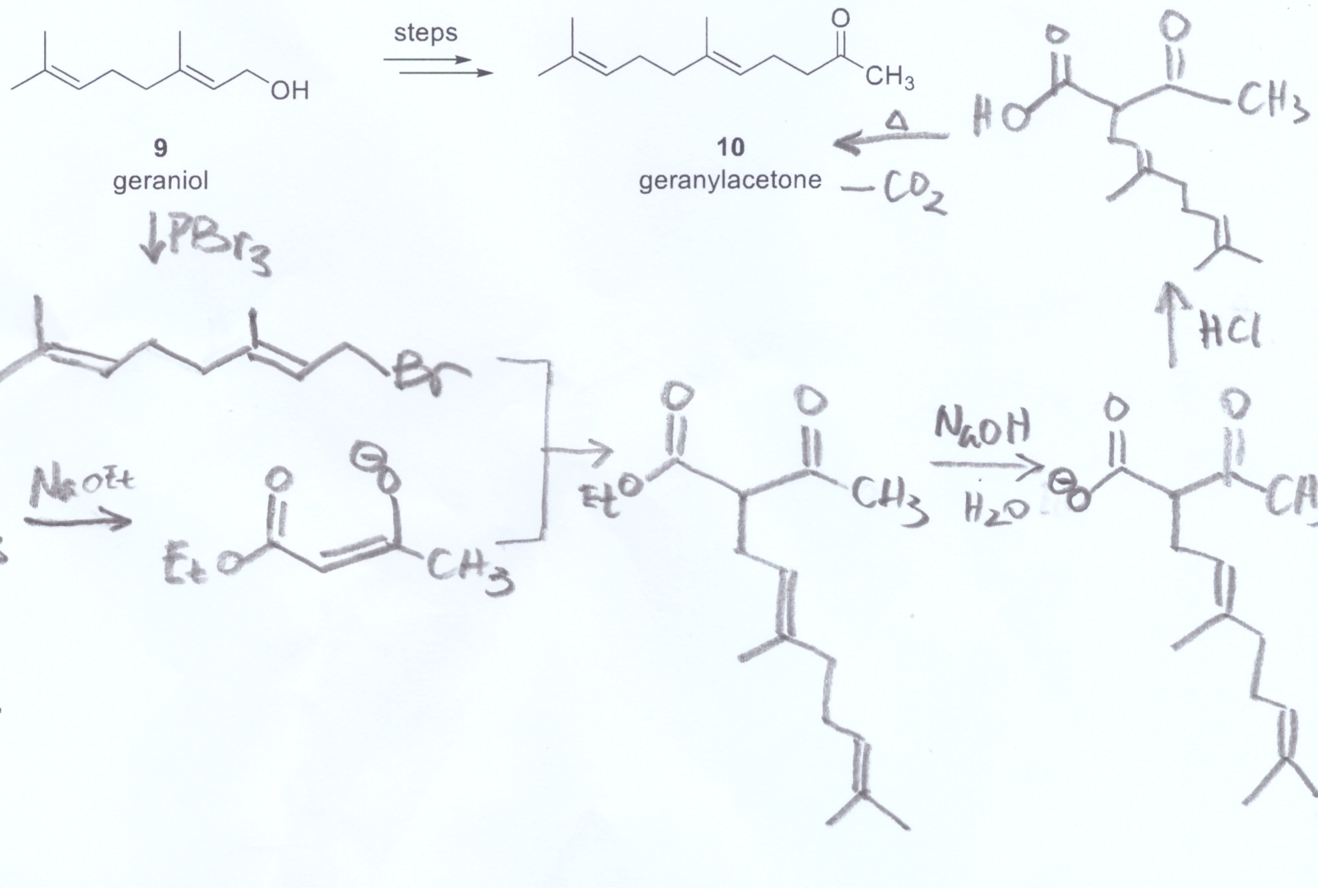


b. Based on the finding in part a, predict the product for the following reaction

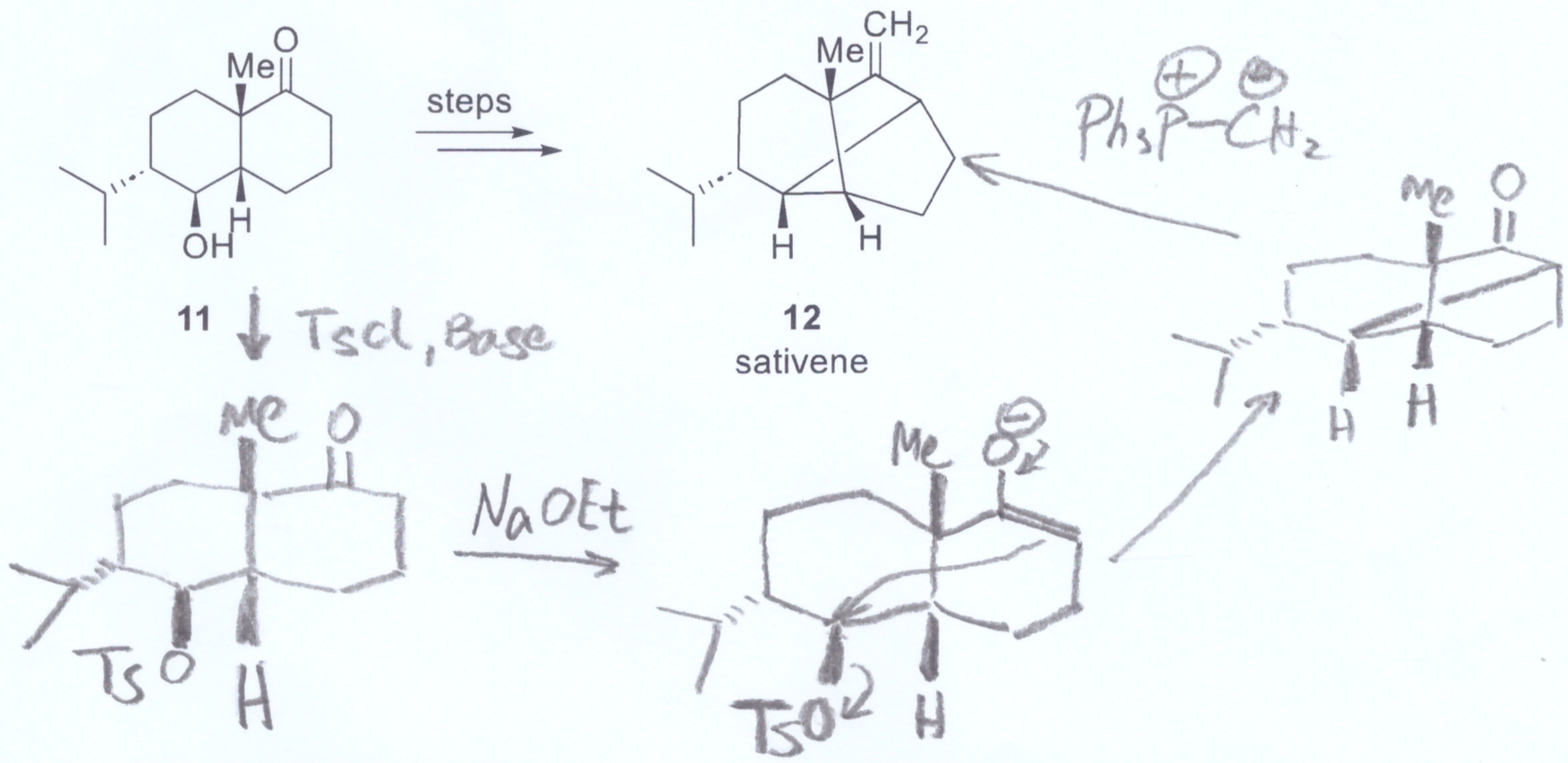


Question 4

a. How could you convert geraniol to geranylacetone? More than 1 step may be required

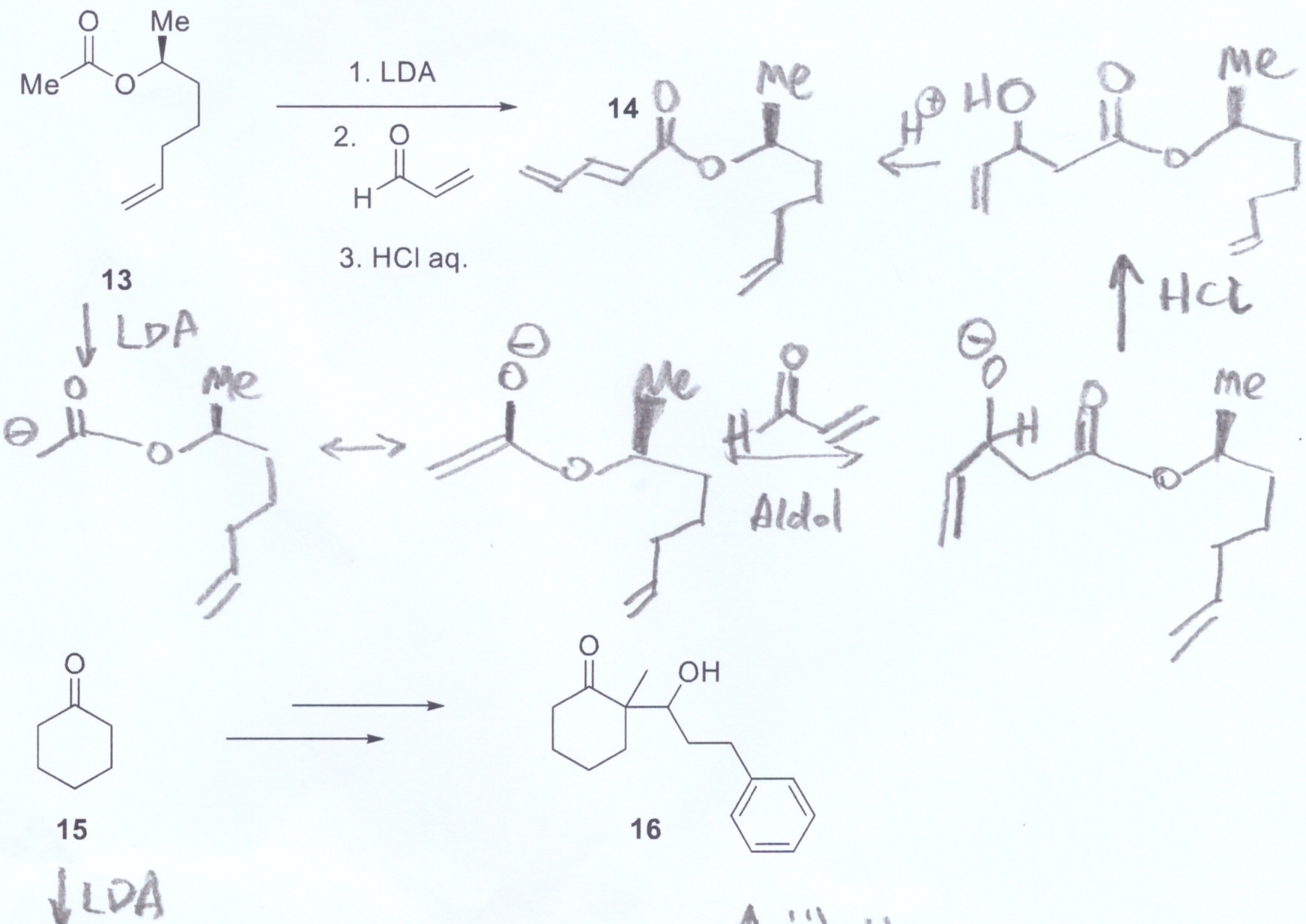


b. Propose a synthesis of a natural product Sativene from ketone **11**



Question 5: Provide the reagents or products for the following transformations

a.



b.

